

10th Grade Learning Guide: Maximax and Maximin

Learning Criterion

C4: Applies the Maximax and Maximin criteria to make optimistic or conservative decisions without probabilities, selecting the alternative with the highest possible payoff or the best worst-case outcome.

All payoffs are in thousands of dollars.

Step 1: Identify

A_1 = Launch dessert line A_2 = Keep current menu
 S_1 = High traffic S_2 = Low traffic

1 Maximax and Maximin

When probabilities are unavailable, decisions can be made by comparing the payoffs for each alternative under each possible state of nature.

	Maximax	Maximin
Focus	Best possible payoff	Worst possible payoff
Attitude	Optimistic	Conservative
First step	Find the maximum in each row	Find the minimum in each row
Final step	Choose the largest maximum	Choose the largest minimum

Step 2: Decide

Maximax

$$M_1 = \max\{42, 8\} = 42$$

$$M_2 = \max\{28, 20\} = 28$$

$$A_{\text{Maximax}}^* = A_1 = \text{Launch dessert line}$$

Maximin

$$m_1 = \min\{42, 8\} = 8$$

$$m_2 = \min\{28, 20\} = 20$$

$$A_{\text{Maximin}}^* = A_2 = \text{Keep current menu}$$

Step 3: Complete the table

Alternative	Max	Min
A_1	42	8
A_2	28	20
A_{Maximax}^*	A_1	
A_{Maximin}^*		A_2

	Alternative	
	A_1	A_2
Max	42	28
Min	8	20
A_{Maximax}^*	A_1	
A_{Maximin}^*		A_2

$$M_i = \max_j P_{ay}(A_i, S_j) \quad A_{\text{Maximax}}^* = \arg \max_i M_i$$

$$m_i = \min_j P_{ay}(A_i, S_j) \quad A_{\text{Maximin}}^* = \arg \max_i m_i$$

2 Solved Example: Coffee Shop

A coffee shop must decide whether to launch a dessert line or keep its current menu. Customer traffic could be high or low.

Alternative	High traffic	Low traffic
Launch dessert line	42	8
Keep current menu	28	20

3 Guided Practice: Clothing Store

A clothing store must decide whether to launch a new advertising campaign or keep its current strategy. Sales could be strong or weak. All payoffs are in thousands of dollars.

Alternative	Strong sales	Weak sales
Launch new campaign	55	5
Keep current strategy	35	22

Step 1: Identify

$$A_1 = \underline{\hspace{2cm}}$$

$$A_2 = \underline{\hspace{2cm}}$$

$$S_1 = \underline{\hspace{2cm}}$$

$$S_2 = \underline{\hspace{2cm}}$$

Step 2: Decide

Maximax

$$M_1 = \max\{\underline{\hspace{1cm}}, \underline{\hspace{1cm}}\} = \underline{\hspace{1cm}}$$

$$M_2 = \max\{\underline{\hspace{1cm}}, \underline{\hspace{1cm}}\} = \underline{\hspace{1cm}}$$

$$A_{\text{Maximax}}^* = \underline{\hspace{2cm}}$$

Maximin

$$m_1 = \min\{\underline{\hspace{1cm}}, \underline{\hspace{1cm}}\} = \underline{\hspace{1cm}}$$

$$m_2 = \min\{\underline{\hspace{1cm}}, \underline{\hspace{1cm}}\} = \underline{\hspace{1cm}}$$

$$A_{\text{Maximin}}^* = \underline{\hspace{2cm}}$$

Step 3: Complete the table

Alternative	Max	Min
A_1	<u> </u>	<u> </u>
A_2	<u> </u>	<u> </u>
A_{Maximax}^*	<u> </u>	<u> </u>
A_{Maximin}^*	<u> </u>	<u> </u>

4 Practice: Restaurant

A restaurant must decide whether to introduce a new lunch menu or keep its current menu. Demand could be high or low. All payoffs are in thousands of dollars.

Alternative	High demand	Low demand
New lunch menu	48	6
Current menu	30	21

Step 1: Identify

$$A_1 = \underline{\hspace{2cm}} \quad A_2 = \underline{\hspace{2cm}}$$

$$S_1 = \underline{\hspace{2cm}} \quad S_2 = \underline{\hspace{2cm}}$$

Step 2: Decide

Maximax

$$M_1 = \max\{\underline{\hspace{1cm}}, \underline{\hspace{1cm}}\} = \underline{\hspace{1cm}}$$

$$M_2 = \max\{\underline{\hspace{1cm}}, \underline{\hspace{1cm}}\} = \underline{\hspace{1cm}}$$

$$A_{\text{Maximax}}^* = \underline{\hspace{2cm}}$$

Maximin

$$m_1 = \min\{\underline{\hspace{1cm}}, \underline{\hspace{1cm}}\} = \underline{\hspace{1cm}}$$

$$m_2 = \min\{\underline{\hspace{1cm}}, \underline{\hspace{1cm}}\} = \underline{\hspace{1cm}}$$

$$A_{\text{Maximin}}^* = \underline{\hspace{2cm}}$$

Step 3: Complete the table

Alternative	Max	Min
A_1	<u> </u>	<u> </u>
A_2	<u> </u>	<u> </u>
A_{Maximax}^*	<u> </u>	<u> </u>
A_{Maximin}^*	<u> </u>	<u> </u>

	A_1	A_2	A_{Maximax}^*	A_{Maximin}^*
Max	<u> </u>	<u> </u>	<u> </u>	<u> </u>
Min	<u> </u>	<u> </u>	<u> </u>	<u> </u>

5 Guided Practice: Technology Company

A technology company must choose a strategy for launching a new device. Market demand could be high, medium, or low. All payoffs are in thousands of dollars.

Alternative	Demand		
	High	Medium	Low
Premium version	80	35	-20
Standard version	60	42	18
Budget version	40	32	25
Delay launch	15	15	15

Step 1: Identify

$$\begin{aligned}
 A_1 &= \underline{\hspace{2cm}} & A_2 &= \underline{\hspace{2cm}} \\
 A_3 &= \underline{\hspace{2cm}} & A_4 &= \underline{\hspace{2cm}} \\
 S_1 &= \underline{\hspace{2cm}} & S_2 &= \underline{\hspace{2cm}} \\
 S_3 &= \underline{\hspace{2cm}}
 \end{aligned}$$

Step 2: Decide

Maximax

$$\begin{aligned}
 M_1 &= \max\{\underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}\} = \underline{\hspace{2cm}} \\
 M_2 &= \max\{\underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}\} = \underline{\hspace{2cm}} \\
 M_3 &= \max\{\underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}\} = \underline{\hspace{2cm}} \\
 M_4 &= \max\{\underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}\} = \underline{\hspace{2cm}} \\
 A_{\text{Maximax}}^* &= \underline{\hspace{2cm}}
 \end{aligned}$$

Maximin

$$\begin{aligned}
 m_1 &= \min\{\underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}\} = \underline{\hspace{2cm}} \\
 m_2 &= \min\{\underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}\} = \underline{\hspace{2cm}} \\
 m_3 &= \min\{\underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}\} = \underline{\hspace{2cm}} \\
 m_4 &= \min\{\underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}\} = \underline{\hspace{2cm}} \\
 A_{\text{Maximin}}^* &= \underline{\hspace{2cm}}
 \end{aligned}$$

Step 3: Complete the table

	A_1	A_2	A_3	A_4	Decision
Max	<u> </u>	<u> </u>	<u> </u>	<u> </u>	$A_{\text{Maximax}}^* = \underline{\hspace{2cm}}$
Min	<u> </u>	<u> </u>	<u> </u>	<u> </u>	$A_{\text{Maximin}}^* = \underline{\hspace{2cm}}$

6 Independent Practice: Sports Equipment Company

A sports equipment company must choose a production strategy for a new product. Demand could be high, medium, or low.

Alternative	Demand		
	High	Medium	Low
Large production	90	30	-15
Medium production	65	45	20
Small production	45	38	28
Maintain production	20	20	20

Step 1: Identify

$$\begin{aligned}
 A_1 &= \underline{\hspace{2cm}} & A_2 &= \underline{\hspace{2cm}} \\
 A_3 &= \underline{\hspace{2cm}} & A_4 &= \underline{\hspace{2cm}} \\
 S_1 &= \underline{\hspace{2cm}} & S_2 &= \underline{\hspace{2cm}} \\
 S_3 &= \underline{\hspace{2cm}}
 \end{aligned}$$

Step 2: Complete the analysis

Alternative	S_1	S_2	S_3	Max	Min
A_1	90	30	-15	<u> </u>	<u> </u>
A_2	65	45	20	<u> </u>	<u> </u>
A_3	45	38	28	<u> </u>	<u> </u>
A_4	20	20	20	<u> </u>	<u> </u>

Maximax

$$A_{\text{Maximax}}^* = \underline{\hspace{2cm}}$$

Maximin

$$A_{\text{Maximin}}^* = \underline{\hspace{2cm}}$$

7 Independent Practice: Bakery Company

A bakery company must choose a production strategy for a new product. Demand could be high, medium, or low.

Alternative	Demand		
	High	Medium	Low
Large production	85	32	-10
Medium production	62	48	18
Small production	44	36	30
Maintain production	22	22	22

Step 1: Identify

$A_1 =$ _____ $A_2 =$ _____
 $A_3 =$ _____ $A_4 =$ _____
 $S_1 =$ _____ $S_2 =$ _____
 $S_3 =$ _____

Step 2: Complete the analysis

Alternative	S_1	S_2	S_3	Max	Min
A_1	85	32	-10	_____	_____
A_2	62	48	18	_____	_____
A_3	44	36	30	_____	_____
A_4	22	22	22	_____	_____

Maximax

$$A_{\text{Maximax}}^* = \text{_____}$$

Maximin

$$A_{\text{Maximin}}^* = \text{_____}$$

8 Independent Practice: Furniture Company

A furniture company must choose a production strategy for a new collection. Demand could be high, medium, or low.

Alternative	Demand		
	High	Medium	Low
Large production	95	28	-18
Medium production	70	50	16
Small production	48	40	32
Maintain production	25	25	25

Step 1: Identify

$A_1 =$ _____ $A_2 =$ _____
 $A_3 =$ _____ $A_4 =$ _____
 $S_1 =$ _____ $S_2 =$ _____
 $S_3 =$ _____

Step 2: Complete the analysis

Alternative	S_1	S_2	S_3	Max	Min
A_1	95	28	-18	_____	_____
A_2	70	50	16	_____	_____
A_3	48	40	32	_____	_____
A_4	25	25	25	_____	_____

Maximax

$$A_{\text{Maximax}}^* = \text{_____}$$

Maximin

$$A_{\text{Maximin}}^* = \text{_____}$$